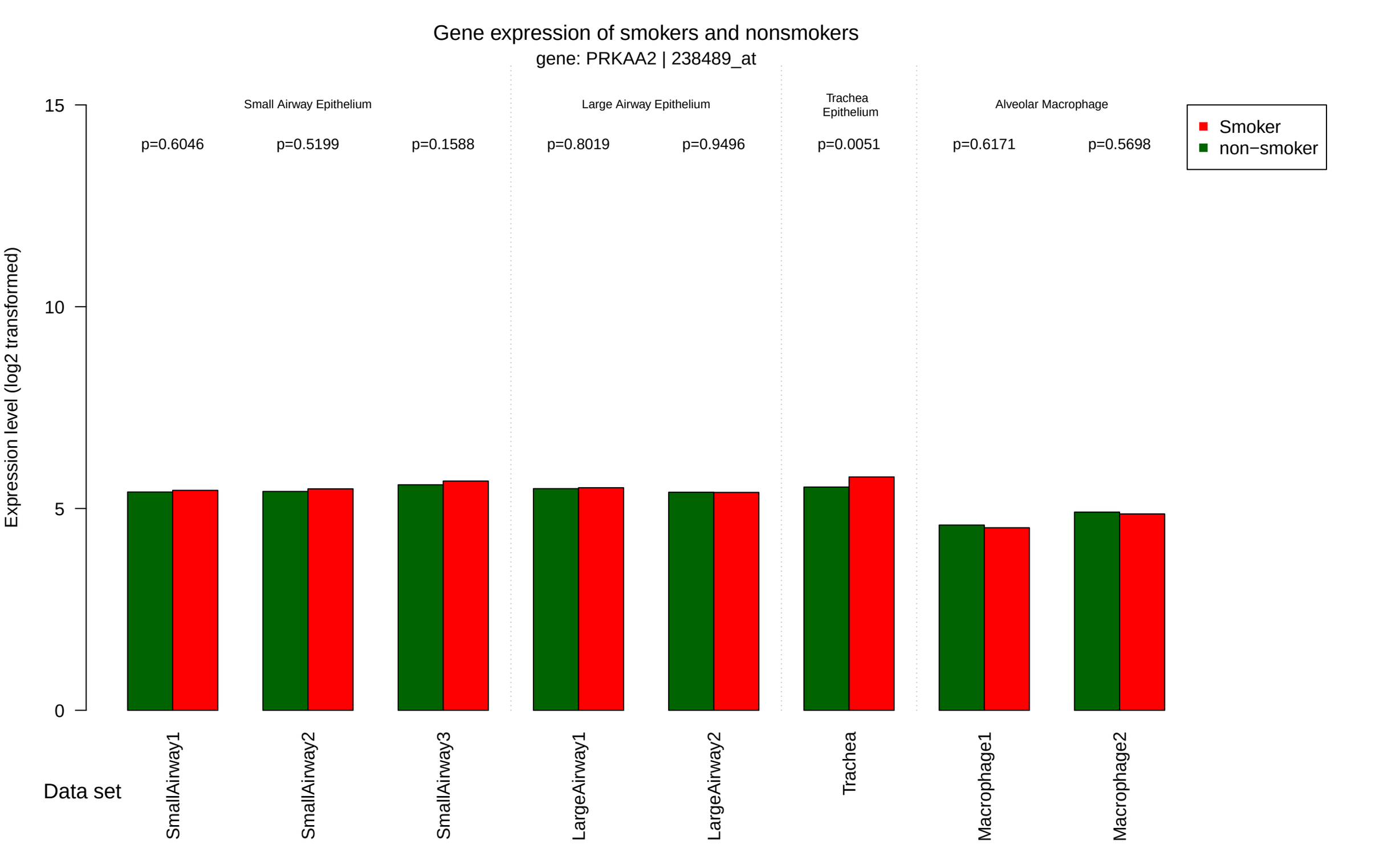
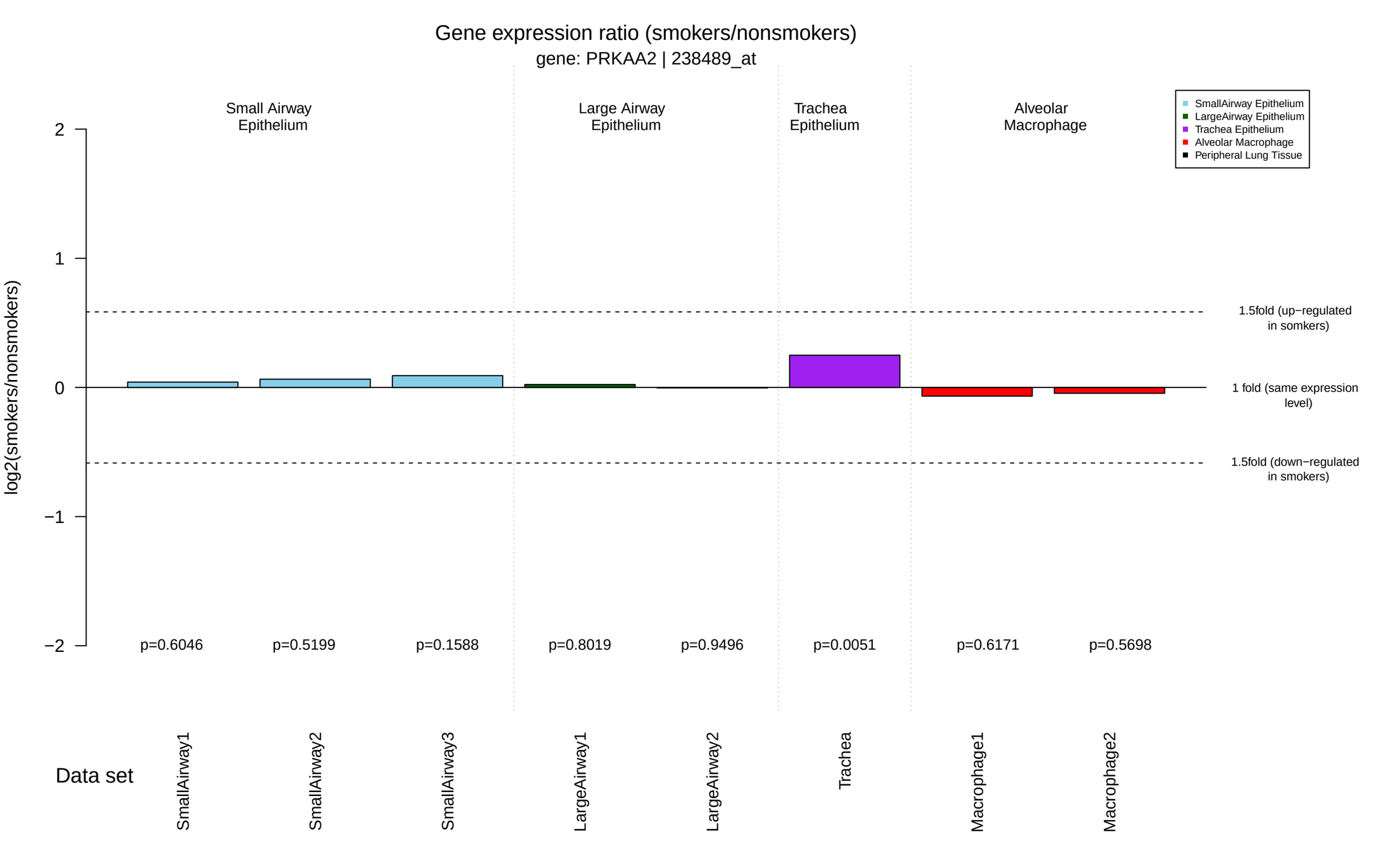
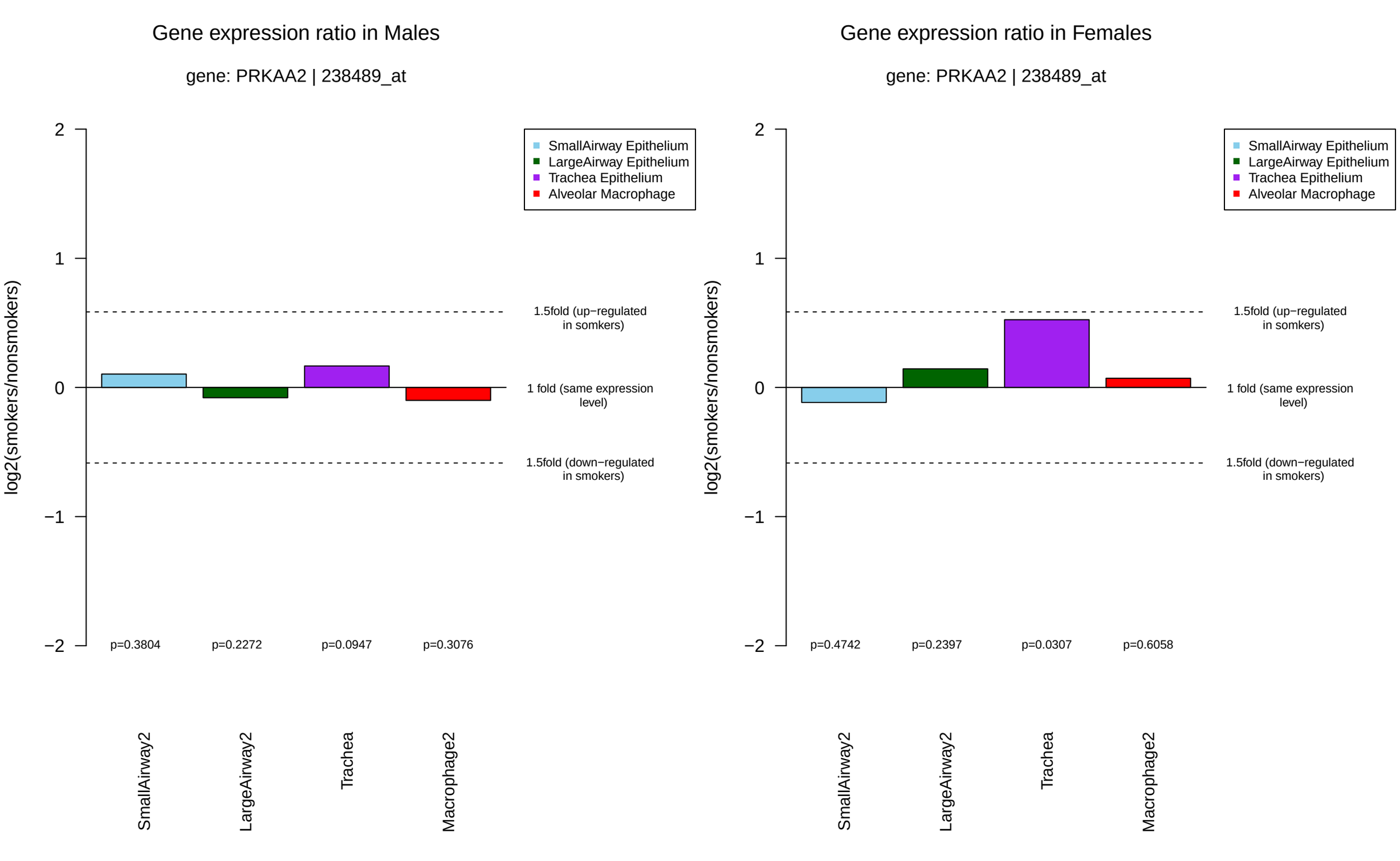
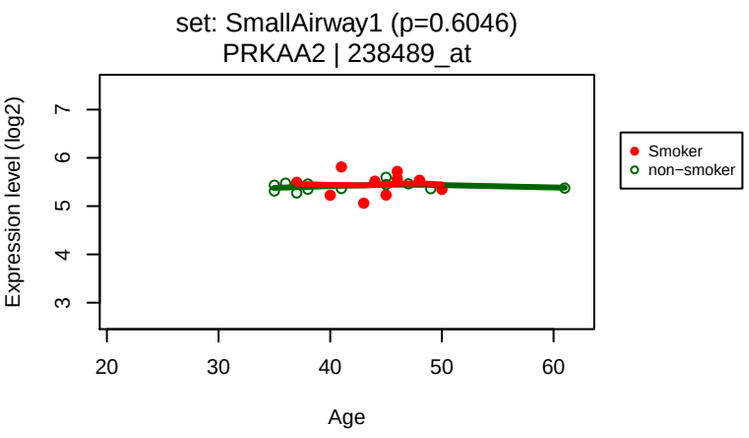
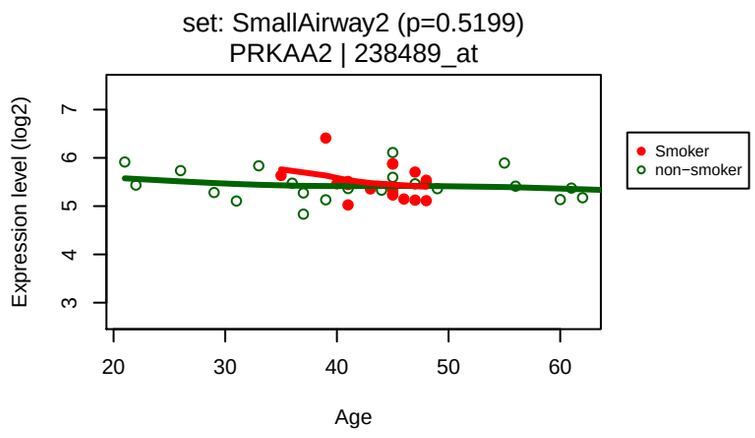
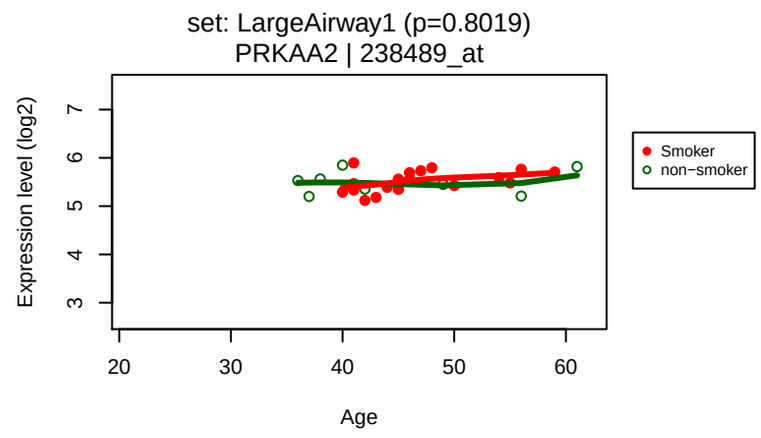

Gene expression of smokers and nonsmokers
gene: PRKAA2 | 238489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6046
p=0.5199
p=0.1588
p=0.8019
p=0.9496
p=0.0051
p=0.6171
p=0.5698
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

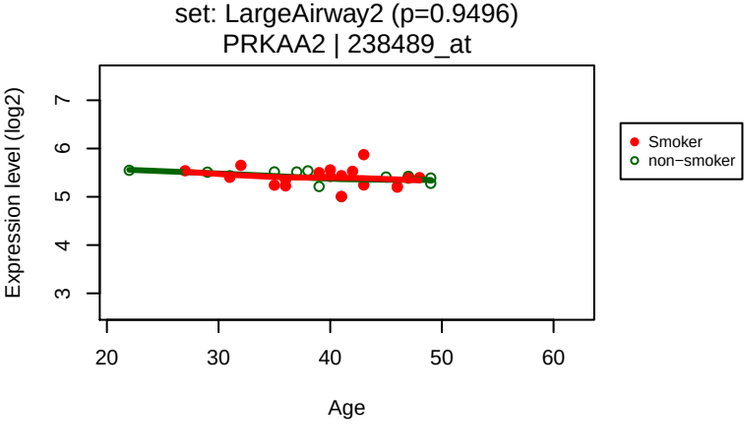

Gene expression ratio (smokers/nonsmokers)
gene: PRKAA2 | 238489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6046
p=0.5199
p=0.1588
p=0.8019
p=0.9496
p=0.0051
p=0.6171
p=0.5698
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

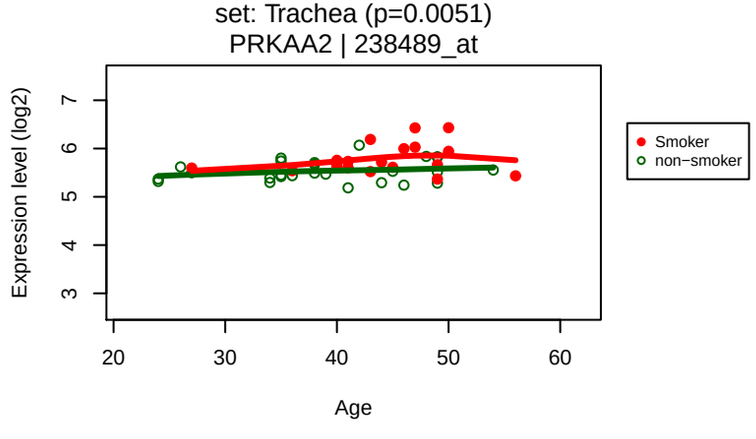

Gene expression ratio in Males
gene: PRKAA2 | 238489_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3804
p=0.2272
p=0.0947
p=0.3076
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAA2 | 238489_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4742
p=0.2397
p=0.0307
p=0.6058
SmallAirway2
LargeAirway2
Trachea
Macrophage2

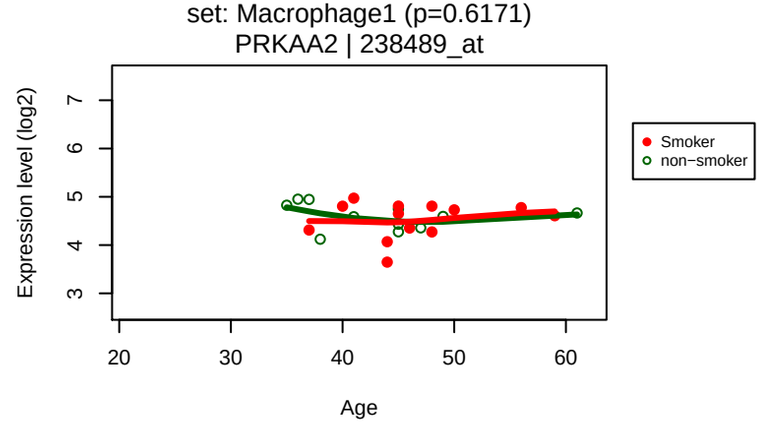

set: SmallAirway1 (p=0.6046)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker

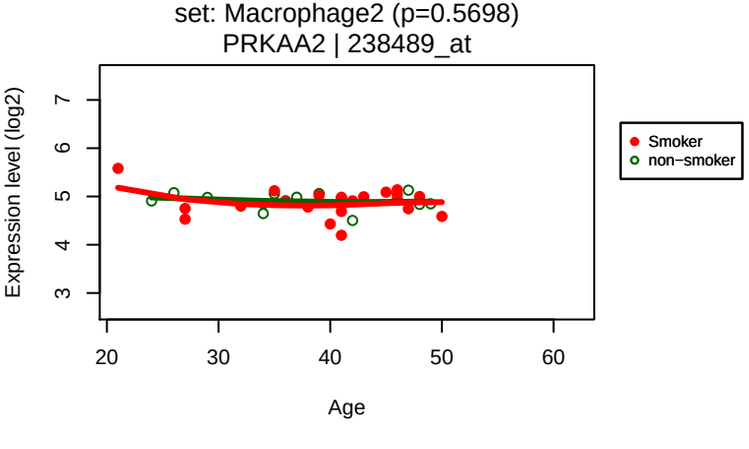

set: SmallAirway2 (p=0.5199)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker

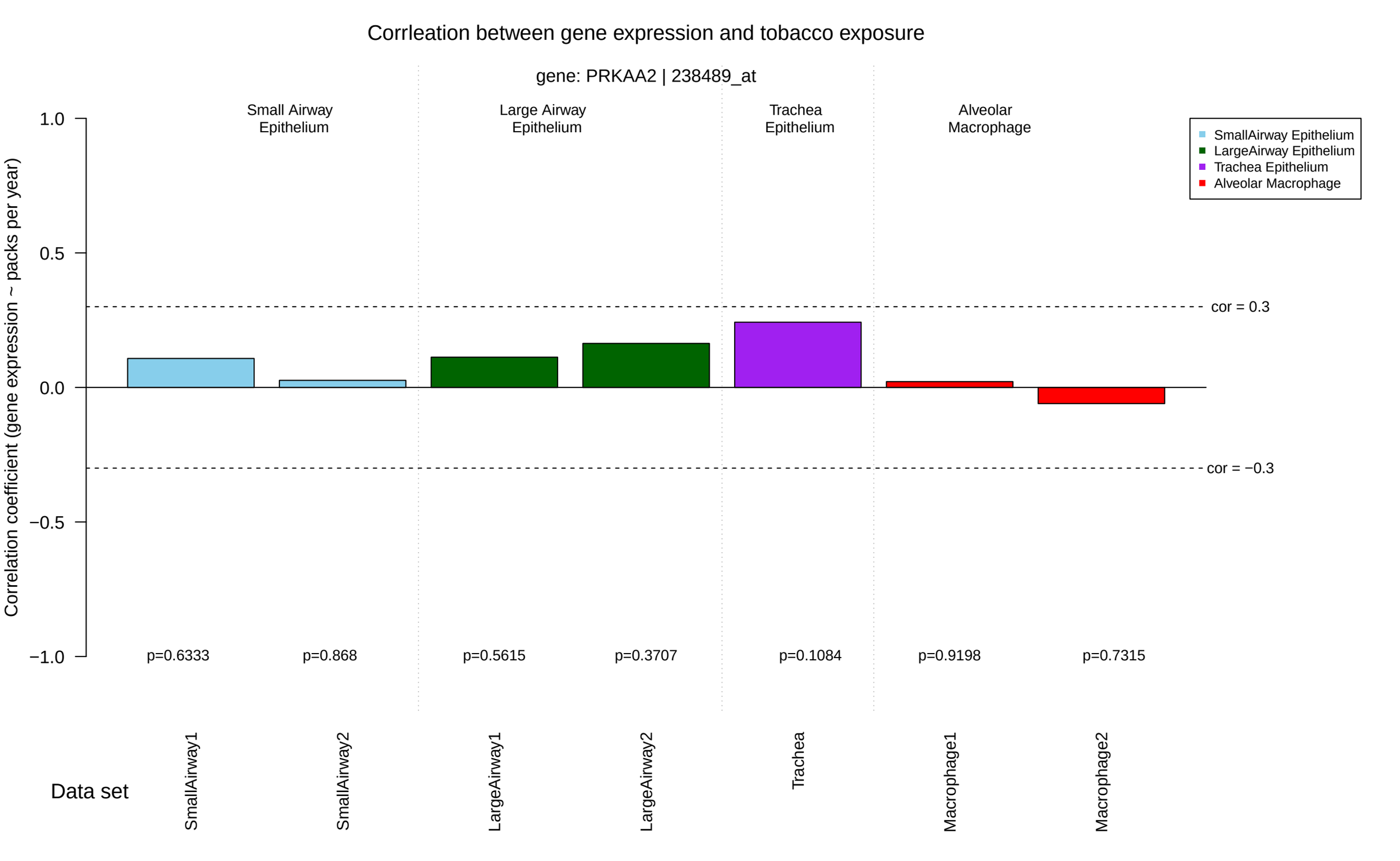

set: LargeAirway1 (p=0.8019)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9496)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0051)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6171)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5698)
PRKAA2 | 238489_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAA2 | 238489_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6333
p=0.868
p=0.5615
p=0.3707
p=0.1084
p=0.9198
p=0.7315
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2